

If you use an online resource to solve any problem, please appropriately cite that source; this includes any AI agent/tool such as ChatGPT, Claude, Gemini, etc.

- a. All hw should be uploaded to canvas as a *pdf*. Make sure that if you scan your handwritten notes that they are legible and appropriately oriented.
- b. Homework is always due on Wednesday after its released at 11:59PM; however, submitting by Monday at 11:59PM will get you extra credit.
- c. All homework will be self-graded, with the exception of one problem (selected uniformly at random) that will be graded.
- d. Self-grading is required to get credit and it is a way to get full credit for your homework (up to any mistakes on the random grading of one problem).
- e. Bonus problems are not required for credit, but they are worth extra credit.

Definitions & Facts. A differentiable function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is **convex** if and only if for all x, y the inequality holds:

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x) \quad (\text{first-order condition}).$$

If f is twice continuously differentiable, a sufficient (and, on open convex domains, also necessary) condition is

$$\nabla^2 f(x) \succeq 0 \quad \text{for all } x \quad (\text{Hessian positive semidefinite}).$$

If $\nabla^2 f(x) \succ 0$ everywhere, then f is *strictly* (indeed, strongly) convex. Recall that a matrix A is positive semidefinite if

$$v^\top A v \geq 0 \quad \forall v \in \mathbb{R}^d,$$

and is positive definite if the inequality is strict. Moreover, as we showed in class, a matrix is positive semidefinite if and only if all its eigenvalues are non-negative.

For a square matrix $A \in \mathbb{R}^{n \times n}$, the **trace** of A , denoted $\text{tr}(A)$, is the sum of its diagonal entries:

$$\text{tr}(A) = \sum_{i=1}^n A_{ii}.$$

Useful properties. For any conformable matrices A, B, C :

$$\begin{aligned} \text{tr}(A + B) &= \text{tr}(A) + \text{tr}(B), \\ \text{tr}(kA) &= k \text{tr}(A) \quad \text{for scalar } k, \\ \text{tr}(A^\top) &= \text{tr}(A), \\ \text{tr}(AB) &= \text{tr}(BA) \quad (\text{cyclic property}), \\ \text{tr}(ABC) &= \text{tr}(BCA) = \text{tr}(CAB). \end{aligned}$$

These identities are widely used when differentiating matrix functions, especially expressions like $\text{tr}(A^\top X)$ or $\text{tr}(X^\top A X)$.

Problems

Problem 1. (Quadratic Function with Linear Term.)

Consider

$$f(x) = \frac{1}{2}x^\top Qx + q^\top x + r, \quad Q = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}, \quad q = \begin{bmatrix} -2 \\ -8 \end{bmatrix}.$$

- (a) Derive expressions for $\nabla f(x)$ and H .
- (b) Find the minimizer x^* .
- (c) Compute $f(x^*)$.
- (d) Sketch or describe the level sets of f . Are they circles or ellipses? Why?

Problem 2. (Deriving $\nabla_X \text{tr}(A^\top X)$ from first principles.)

Let $A, X \in \mathbb{R}^{m \times n}$ and define

$$f(X) = \text{tr}(A^\top X).$$

- (a.) Write $f(X)$ out as a double sum over entries of A and X .
- (b.) Compute the partial derivative $\frac{\partial f}{\partial X_{ij}}$ directly from that expression.
- (c.) Assemble your results into a matrix to find $\nabla_X f(X)$.
- (d.) Verify your result by checking a simple case where $m = n = 2$ and A and X have arbitrary entries.
- (e.) Use the result to find $\nabla_X \text{tr}(X^\top A)$.

Problem 3. (Matrix derivatives.)

Throughout, use the rule that $\nabla_X \text{tr}(A^\top X) = A$ and linearity of differentiation. All variables are real-valued matrices.

- (a) Let $f(X) = \text{tr}(A^\top X)$. Derive an expression for $\nabla_X f(X)$.
- (b) Let $f(X) = \text{tr}(X^\top A)$. Derive an expression for $\nabla_X f(X)$.
- (c) Let $f(X) = \text{tr}(AXB)$. Derive an expression for $\nabla_X f(X)$.
- (d) Let $f(X) = \text{tr}(AX^\top B)$. Derive an expression for $\nabla_X f(X)$.

Problem 4. (Least Angle Property of Least Squares.) Suppose the matrix $X \in \mathbb{R}^{m \times n}$ has linearly independent columns, and let $y \in \mathbb{R}^m$. Let $\hat{\theta}$ denote the least-squares approximate solution of $X\theta = y$, i.e.

$$(X^\top X)\hat{\theta} = X^\top y.$$

We will establish several geometric properties of the least-squares solution.

- a. **Inner product property.** Show that for any $\theta \in \mathbb{R}^n$,

$$(X\theta)^\top y = (X\theta)^\top (X\hat{\theta}),$$

i.e., the inner product of $X\theta$ and y is the same as that of $X\theta$ and $X\hat{\theta}$. (Hint: use $(X\theta)^\top y = \theta^\top (X^\top y)$ and $(X^\top X)\hat{\theta} = X^\top y$.)

- b. **Angle between $X\hat{\theta}$ and y .** Show that when both $X\hat{\theta}$ and y are nonzero,

$$\frac{(X\hat{\theta})^\top y}{\|X\hat{\theta}\| \|y\|} = \frac{\|X\hat{\theta}\|}{\|y\|}.$$

The left-hand side is the cosine of the angle between $X\hat{\theta}$ and y . Hence $X\hat{\theta}$ and y form the smallest possible angle.

- c. **Least angle property of least squares.** The choice $\theta = \hat{\theta}$ minimizes the distance between $X\theta$ and y . Show that $\theta = \hat{\theta}$ also minimizes the angle between $X\theta$ and y . (You may assume $X\theta$ and y are nonzero.) *Remark.* For any scalar $\alpha > 0$, $\theta = \alpha\hat{\theta}$ also minimizes the angle between $X\theta$ and y .

Problem 5. (Linear Functions (8.6 VMLS).) For each description of y , express it as $y = Ax$ for some A .

- (a) y_i is the difference between x_i and the average of $x_1, \dots, x_{i-1}$ (with $y_1 = x_1$).
- (b) y_i is the difference between x_i and the average of all the other entries, i.e., the average of $\{x_j : j \neq i\}$.

Problem 6. (Weighted least squares.) Let $W \in \mathbb{R}^{n \times n}$ be a symmetric positive definite matrix and define

$$f(\theta) = (X\theta - y)^\top W(X\theta - y).$$

- (a) Show that $\nabla_\theta f(\theta) = 2X^\top W(X\theta - y)$.
- (b) Derive the normal equations—i.e., c, d such that $d = X^\top c$. Under what conditions can you find an expression for the minimizer θ^* , and what is the expression?
- (c) Explain in words how the weighting matrix W affects which data points have more influence.

Problem 7. (Composition rule practice.) Let $f(x) = g(Ax + b)$ where $g : \mathbb{R}^m \rightarrow \mathbb{R}$ is differentiable and $A \in \mathbb{R}^{m \times n}$.

- (a) Show that $\nabla_x f(x) = A^\top \nabla g(Ax + b)$.
- (b) If $g(z) = \frac{1}{2} \|z\|_2^2$, compute $\nabla_x f(x)$ and $\nabla_x^2 f(x)$ explicitly.
- (c) Explain why this captures the gradient and Hessian of $\frac{1}{2} \|Ax + b\|_2^2$ without expanding coordinates.